Mart Reference

Citi Bike (NYC) vs Divvy (Chicago): the 16 CSV marts and the Tableau story

Data accessed 13 October 2026

Purpose. This reference describes the 16 CSV marts in `data/marts/`: what each file contains, which tab of the 4-tab Tableau story uses it (New York, NYC vs Chicago, Weather, Stations), and what to state alongside it. The packaged workbook `tableau/citibike_story.twbx` is built from these files by `tableau/build_story.py`; files not used by a tab are there for analysis in any BI tool.

Where: folder `data/marts/` in this repository. Total size is about 1 MB; the largest file has 4,384 rows.
Period: July 2023 – June 2026 (36 months).
Findings behind the files: `docs/findings.md`.

1 The Files at a Glance

File	Rows	Dashboard tab	Shows
`daily_weather_ridership`	4,384	all except Stations	every day, both cities, members vs. casual, with weather
`weather_rain_effect`	16	– (Weather tab recomputes it)	trips vs. a normal day, by rain level
`weather_snow_effect`	8	– (Weather tab recomputes it)	snow days vs. other winter days
`weather_temperature_effect`	20	–	average trips by temperature band
`city_monthly_engagement`	72	–	monthly trips and member share, both cities
`seasonality`	12	– (Weather tab recomputes it)	winter riding as a share of summer
`city_comparison`	6	NYC vs Chicago (reshaped)	member share and trip length per year
`price_history`	27	NYC vs Chicago	every plan price over time, with source link
`price_breakeven`	6	NYC vs Chicago	day passes that equal one annual membership
`nyc_monthly_kpis`	36	New York	members, sign-ups, revenue, MRR, support
`nyc_revenue_mix`	288	New York	revenue by category per month
`survey_plan_held`	8	–	which plan Citi Bike users used
`survey_frequency`	6	–	how often people use Citi Bike
`survey_non_use_reasons`	8	New York	why people do not use Citi Bike
`station_activity`	3,544	Stations	every station with location and trips
`ebike_share`	16	NYC vs Chicago	e-bike share by city and rider type

2 Files by Topic

2.1 Weather (Hypotheses 1 and 2; Weather tab)

`daily_weather_ridership`: the main file. One row per day, city and rider type.

- Key columns: `trips`; `trips_vs_baseline` (1.00 = a normal day for that city, month and weekday type; 0.60 = 40% fewer trips); weather columns `tmax_c`, `precip_mm`, `snow_mm`, `wind_ms`; ready-made groups `rain_band`, `temp_band`, `snow_day`.
- The Weather tab computes its rain, snow and winter figures from this file, so they follow the dashboard filters.

`weather_rain_effect`, `weather_snow_effect`: the same comparisons as fixed tables. Example: heavy rain in NYC means casual trips -44% and member trips -37% vs. a dry day (both on the normal-day ratio).
`weather_temperature_effect`: average trips by temperature band (this includes the season effect).
`seasonality`: `winter_to_summer_ratio`. Chicago casual riders keep only 10–13% of their summer riding in winter; NYC members keep 39–57%.

2.2 Cities and Prices (Hypotheses 3 and 4; NYC vs Chicago tab)

`city_comparison`: member share (NYC 81–83%, Chicago 64–65%) and trip length per ride year; the workbook uses a long (reshaped) version of it.
`city_monthly_engagement`: monthly trips and member share per city.
`price_history`: the annual price over time (NYC \$205 $\rightarrow$ \$219.99 $\rightarrow$ \$239; Chicago \$143.90). Each row includes its `source_url`.
`price_breakeven`: an NYC membership costs the same as 9.6–11.6 day passes; Chicago's costs 7.2–8.0.
`ebike_share`: `ebike_share` by city and rider type (NYC casual 77.5% vs. member 70.5%; Chicago 71.2% vs. 65.7%).

2.3 NYC Business KPIs (all five KPI categories; New York tab)

`nyc_monthly_kpis`: one row per month.

- Acquisition: `annual_signups_or_renewals`, `single_trip_passes`, `day_passes`
- Engagement: `trips_per_member`
- Retention: `total_annual_members`, `net_change_mom`, `net_growth_yoy`
- Monetization: `total_revenue`, `revenue_before_tax`, `membership_revenue`, `lyft_pink_revenue`, `rider_revenue`, `rider_revenue_per_trip` (all rider revenue, including memberships and Lyft Pink, excluding tax and sponsorship, divided by all trips), `list_price_mrr`, `list_price_arr`
- Satisfaction: `calls_per_100_trips`, `avg_handle_minutes`

`nyc_revenue_mix`: revenue by `category` and month (36 months $\times$ 8 categories). E-bike fees are the largest revenue line. Tax is left out (it is sales tax collected for the government, not revenue); `share_of_total` is the share of revenue before tax. The derived May 2026 sponsorship value ($-$\$384k) comes from the year-to-date method, which assigns a later restatement of an earlier month to the missing month.

2.4 Customers (NYC DOT survey 2024)

`survey_plan_held`: `weighted_share` by plan. 44% of Citi Bike users hold a subscription.
`survey_frequency`: how often people ride.
`survey_non_use_reasons`: "too expensive" is cited by 35%, "not available nearby" by 28% (New York

tab).
Always use the weighted share columns; `respondents_n` shows the sample size.

2.5 Stations (Stations tab)

`station_activity`: map using `lat`/`lng`; circle size = member or casual trips (chosen with a rider-type control); `member_share` is also available. Data comes from the four sample months.

3 Why These Files Are Good to Use

1. **Real and validated.** Every value comes from an official source: Citi Bike, Divvy, the U.S. Bureau of Transportation Statistics, NOAA and NYC DOT. Trip counts match the federal statistics within 0.1%, and a separate source-by-source audit checked report figures against the original PDFs.
2. **Small and fast.** All files are far below Tableau Public's row limit, so dashboards load instantly.
3. **Ready to use.** One row per day, month or station, with clear column names. No cleaning is needed in Tableau.
4. **Fair comparisons are built in.** The hard calculations were done in SQL. For example, `trips_vs_baseline` removes season and weekday effects, so the weather comparison is like-for-like.
5. **Filter-friendly.** Shared columns (`city_id`, `rider_type`, `year_month`, `season`) let one filter control a whole dashboard.
6. **Traceable.** Each file is produced by a SQL view in `sql/kpis/`, so every number can be explained step by step.

4 What to State on the Dashboard

- Weather and city results are **associations, not causes**.
- `weather_temperature_effect` includes the season effect (cold days are winter days). Use the rain and snow files for weather effects.
- `nyc_monthly_kpis`: **Feb and May 2026**: reports not published, so members, sign-ups and support figures are blank; revenue is calculated from year-to-date totals and `revenue_basis` = `derived_from_ytd` marks it. `report_scope_changed` marks March 2025, when Jersey City trips were added to the reports. MRR is an upper bound because some members pay discounted rates.
- No churn is shown: the reports combine new sign-ups and renewals.
- Survey files cover **people who cycle** (about 34% of adults).
- Membership, revenue and support data exist for **NYC only**.
- Credit the sources on the dashboard: Citi Bike, Divvy, U.S. BTS, NOAA, NYC DOT.

5 Using the Files in Tableau Public

1. Open Tableau Public → *Connect* → *Text file* → choose a CSV from `data/marts/`.
2. Use one data source per sheet or dashboard. Add the others via *Data* → *New Data Source*; you do not need to join the files.
3. Check field types after loading: `date` as Date, `lat`/`lng` as Latitude/Longitude, shares as percentages.
4. **Suggested first chart:** `daily_weather_ridership`, `tmax_c` on columns, `trips_vs_baseline` on rows, colour by `rider_type`, filter by `city_id`.